

2024, Mar. 2024 Release LOC:Acute Adult**TIA**

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview**Select Day**

Initial review ^(1, 2, 3)

Episode Day 1 ^(1, 2)

Episode Day 2 ⁽¹⁰⁾

Episode Day 3 ⁽¹⁶⁾

Discharge Screens ⁽¹⁷⁾**Utilization Benchmarks****Notes**

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Overview**Introduction:**

A transient ischemic attack (TIA) is a neurologic condition associated with a risk of stroke. It is a brief episode of neurologic dysfunction that begins suddenly and resolves rapidly and completely, with the vast majority of TIAs lasting minutes to less than an hour. It is caused by focal brain, spinal cord, or retinal ischemia without evidence of acute infarction on neuroimaging.

TIA is a clinical syndrome that can be challenging because its symptoms are transient and highly variable. Imaging criteria coupled with risk assessment tools which are being further refined and tested help with diagnosis. Symptoms may include hemiparesis, hemiparesthesia, hemineglect, visual field deficit, gaze deviation and eye movement abnormalities, dysarthria, aphasia, or gait and balance disturbances.

Rapid medical evaluation is essential for TIA management and potential stroke prevention. Patients considered at high risk for a stroke include those with conditions associated with an embolic source, such as atrial fibrillation, cardiomyopathy, or valvulopathy; known carotid stenosis; a previous stroke; and/or crescendo TIA (Mendelson and Prabhakaran, *Jama* 2021, 325: 1088-98; Cereda and Olivot, *Current atherosclerosis reports* 2018, 20: 56; Gomez et al., *F1000Res* 2017, 6: 1893; Lo et al., *Critical Issues in the Evaluation of Adults With Suspected TIA in the ED*. In: ACEP; 2016).

Evaluation and Treatment:

Initial evaluation involves history and physical, electrocardiogram (ECG), and a basic laboratory assessment. Imaging studies are done to rule out transient ischemic attack (TIA) mimics or alternative diagnoses and to search for potential embolic sources and cerebrovascular disease. TIA mimics can include migraines, seizures, syncope, or functional or anxiety related sensory symptoms.

Diffusion-weighted imaging (DWI), a form of magnetic resonance imaging (MRI), and cervical vascular imaging have been shown to be useful short-term risk predictors for the development of stroke in patients presenting with TIA (Mendelson and Prabhakaran, *Jama* 2021, 325: 1088-98; Lo et al., *Critical Issues in the Evaluation of Adults With Suspected TIA in the ED*. In: ACEP; 2016).

The initial workup may include:

- Brain imaging: computer tomography (CT), MRI, or DWI
- Vascular imaging: computer tomography arteriogram (CTA), magnetic resonance arteriogram (MRA), or carotid ultrasonography
- Cardiac imaging: transthoracic echocardiogram (TTE), transesophageal echocardiogram (TEE), or multidetector cardiac computer tomography

Current treatment addresses the underlying cause of the TIA. Guidelines recommend antiplatelet agents or anticoagulants and the treatment of hypertension, hyperlipidemia, and arterial obstruction. Warfarin anticoagulation is appropriate for patients with valvular atrial fibrillation. Non-vitamin K oral anticoagulants (NOACs), such as dabigatran, rivoraxaban, apixaban and edoxaban, have been shown to reduce the risk of TIA in patients with underlying non-valvular atrial fibrillation. Statin therapy with intensive lipid-lowering effects is recommended for all patients with ischemic stroke and transient ischemic attack to reduce the risk of stroke and cardiovascular event (Cereda and Olivot, Current atherosclerosis reports 2018, 20: 56; Gomez et al., F1000Res 2017, 6: 1893; Rothwell et al., Lancet 2016, 388: 365-75; Amarenco et al., N Engl J Med 2016, 374: 1533-42; Kernan et al., Stroke 2014, 45: 2160-236). *Many TIA patients are at high risk for readmission, and careful attention to discharge planning is essential.*

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Mendelson and Prabhakaran, Jama 2021, 325: 1088-98; Cereda and Olivot, Current atherosclerosis reports 2018, 20: 56; Gomez et al., F1000Res 2017, 6: 1893; Lo et al., Critical Issues in the Evaluation of Adults With Suspected TIA in the ED. In: ACEP; 2016; Kernan et al., Stroke 2014, 45: 2160-236).

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽⁴⁾– Neurological deficit resolved or resolving ⁽⁵⁾– CT or MRI performed ⁽⁶⁾– Vascular imaging of carotid artery performed within 24h ⁽⁷⁾● **Episode Day 1, One:** ^(1, 2)

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽⁴⁾– Neurological deficit resolved or resolving ⁽⁵⁾– CT or MRI performed ⁽⁶⁾– Vascular imaging of carotid artery performed within 24h ⁽⁷⁾– Antiplatelet agent or anticoagulant (includes PO) ^(8, 9)● **Episode Day 2, One:** ⁽¹⁰⁾● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽¹¹⁾– Neurological stability ⁽¹²⁾– Systolic blood pressure within acceptable limits ^(13, 14)● **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**● **Hypertension, Both:**

– Systolic blood pressure > 180 mmHg

– Antihypertensive medication (includes PO) ⁽¹⁵⁾● **Episode Day 3, One:** ⁽¹⁶⁾● **OBSERVATION, One:**● **Responder**, discharge expected today if clinically stable last 12h, **Both:** ⁽¹¹⁾– Neurological stability ⁽¹²⁾– Systolic blood pressure within acceptable limits ^(13, 14)● **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**

– For a condition other than TIA use appropriate subset (Episode Day 1) based on clinical findings

– Refer for secondary review

Discharge, Level of Care, ≥ One: ⁽¹⁷⁾● **HOME, Both:**● **Level of care appropriateness, Both:**– Home environment safe and accessible ⁽¹⁸⁾

– Patient or caregiver demonstrates ability to manage care

● **TIA discharge planning, All:** ⁽¹⁹⁾– Provide comprehensive written discharge and teaching instructions ⁽²⁰⁾● **Education on the importance of, All:**

– Strict medication adherence

– Statin therapy ^(21, 22)

– Follow-up care

– Smoking cessation and avoiding passive smoke

– Eliminating or reducing alcohol intake

– Engaging in at least 30 minutes of physical activity daily

– Achieving and maintaining a healthy weight

– Understanding when to seek help

– Medication reconciliation ⁽²³⁾– Follow-up care planned ⁽²⁴⁾

- Identify and address transportation needs
- **HOME CARE, One:**
 - Level of care appropriateness, **All:**
 - Home environment safe and accessible ⁽¹⁸⁾
 - Patient or caregiver willing and able to learn care
 - TIA discharge planning, **All:** ⁽¹⁹⁾
 - Provide comprehensive written discharge and teaching instructions ⁽²⁰⁾
 - Education on the importance of, **All:**
 - Strict medication adherence ⁽²²⁾
 - Statin therapy ⁽²¹⁾
 - Follow-up care
 - Smoking cessation and avoiding passive smoke
 - Eliminating or reducing alcohol intake
 - Engaging in at least 30 minutes of physical activity daily
 - Achieving and maintaining a healthy weight
 - Understanding when to seek help
 - Medication reconciliation ⁽²³⁾
 - Follow-up care planned ⁽²⁴⁾
 - Identify and address transportation needs
 - Outpatient management contraindicated or unavailable ^(25, 26)
 - Skilled services, ≥ **One:**
 - Skilled nursing ⁽²⁷⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽²⁸⁾
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽²⁹⁾
 - Skilled treatment, ≥ **One:**
 - Clinical assessment ⁽³⁰⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽³¹⁾
- **SKILLED THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ⁽³²⁾
 - Functional impairments requiring at least supervision ⁽³³⁾
 - Able to tolerate 1-2h/d of skilled therapy ≥ 5d/wk
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽²³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ⁽³²⁾
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽³³⁾
 - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽²³⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ⁽³²⁾
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽³³⁾
 - Able to tolerate 2-3h/d of skilled therapy $\geq 5d/wk$ and ≥ 2 therapy disciplines
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽²³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **ACUTE REHAB, Both:**
 - Level of care appropriateness, **Both:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3x/wk$
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ⁽³²⁾
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽³³⁾
 - Able to tolerate $\geq 15h/wk$ of skilled therapy and ≥ 2 therapy disciplines ⁽³⁴⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽²³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
 - Level of care appropriateness, **Both:**
 - Treatment precluded in a lower level
 - In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽³⁵⁾
 - Ventilator dependent $\geq 6h/d$ and weaning planned ^(36, 37)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽³⁸⁾
- Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽²³⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
069 TRANSIENT ISCHEMIA WITHOUT THROMBOLYTIC	n/a	2.1	CMS GMLOS
TRANSIENT ISCHEMIC ATTACK (TIA)	64%	2.2	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category,

and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Transition plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. There is a high risk of early and long-term hospital readmissions following a transient ischemic attack (TIA) and acute ischemic stroke. Following TIA and acute stroke, 17.4%(0.174) of patients will require readmission within the first month, and 42.5%(0.425) within the first year. Some factors associated with readmission include acute cerebrovascular disease, infection, occlusion or stenosis of pre-cerebral arteries, and acute renal failure (Yousufuddin et al., Journal of Stroke and Cerebrovascular Diseases 2019, 28: 167-74; Bambhroliya et al., JAMA network open 2018, 1: e181190-e; Vahidy et al., Stroke 2017, 48: 1386-8; Zhong et al., Neurological Sciences 2016: 1-8).

3:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a

decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

4:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

5:

A transient ischemic attack (TIA) is neurological syndrome associated with a risk of stroke. It is a brief episode of neurologic dysfunction that begins suddenly and resolves rapidly and completely, with the vast majority of TIAs lasting minutes to less than an hour.

6:

Diffusion-weighted imaging (DWI), along with cervical vascular imaging, has been shown to predict the short-term risk for stroke in patients presenting with a transient ischemic attack (TIA) (Cereda and Olivot, Current atherosclerosis reports 2018, 20: 56; Lo et al., Critical Issues in the Evaluation of Adults With Suspected TIA in the ED. In: ACEP; 2016).

7:

Vascular imaging with computed tomography angiography (CTA), magnetic resonance angiography (MRA), or carotid ultrasonography is recommended early after diagnosis of transient ischemic attack to determine patients at high risk of stroke who may benefit from carotid revascularization (Kleindorfer et al., Stroke 2021, 52: e364-e467).

8:

Studies combining clopidogrel and aspirin have been shown to decrease the risk of ischemic events compared to aspirin alone in patients who have experienced minor strokes or high risk transient ischemic attack. This combination therapy does, however carry an increase in the risk of major and minor hemorrhage (SC et al., N Engl J Med 2018, 379: 215-25).

9:

Guidelines strongly recommend oral anticoagulants for patients with atrial fibrillation (AF). Non-vitamin K antagonist oral anticoagulants (NOACs), such as dabigatran, rivoraxaban, apixaban, and edoxaban, are recommended over warfarin in NOAC eligible patients with AF (e.g., those without contraindications to specific medications and without moderate to severe mitral stenosis or a mechanical heart valve). Warfarin anticoagulation is appropriate for patients with valvular AF. NOACs have been shown to reduce the risk of TIA or stroke in patients with underlying non-valvular atrial fibrillation (Kleindorfer et al., Stroke 2021, 52: e364-e467; Beaser and Cifu, Jama 2019, 321: 1100-1).

10:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level

services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

11:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

12:

Neurological stability is a clinical state characterized by:

- Lack of deterioration or return to baseline in mental status or level of consciousness
- Seizures absent or controlled
- Absence of neurological deficit
- Stabilization of neurological deficit that develops during current hospitalization

13:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

14:

Secondary prevention for stroke or transient ischemic attack (TIA) includes management of hypertension. For patients experiencing a stroke or TIA that were not previously receiving antihypertensive medication therapy, initiation is suggested for systolic blood pressure (SBP) greater than or equal to 140 mmHg, or a diastolic blood pressure (DBP) greater than or equal to 90mmHg. Antihypertensive therapy should be resumed for those patients previously receiving drug therapy in order to prevent recurrent stroke and other vascular events.

The following lifestyle alterations are helpful in the reduction of blood pressure:

- Salt restriction
- Weight loss
- Increase fruit and vegetables in the diet
- Decrease fat intake
- Limit alcohol intake
- Regular physical activity
- Smoking cessation

Goals for a target blood pressure should be individualized. However, managing the blood pressure to a SBP less than 140 mmHg and DBP less than 90 mmHg during the recovery period is recommended (Kleindorfer et al., Stroke 2021, 52: e364-e467; Powers et al., Stroke 2019, 50: e344-e418).

15:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

16:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-**

based services and supported by clinical documentation would be appropriate for inpatient status; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

17:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

18:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

19:

The following discharge instructions may be individualized as appropriate (Gomez et al., F1000Res 2017, 6: 1893; Winstein et al., Stroke 2016, 47: e98-e169; Kernan et al., Stroke 2014, 45: 2160-236):

- Call emergency medical services immediately if there is any sudden weakness or numbness of the face, arm, or leg; blurry or decreased vision; inability to speak or understand; dizziness, loss of balance or falling; severe headache
- Take all medications as prescribed
- Eat a well-balanced diet low in cholesterol and saturated fat; drink 6-8 glasses of water or juice every day to maintain good hydration and avoid constipation
- Have blood pressure checked regularly as uncontrolled high blood pressure is a risk factor for stroke
- Diabetics should maintain tight control of their blood glucose since elevated blood glucose contribute to stroke
- Consume alcohol in moderation
- Smokers should consider participation in a smoking cessation program

20:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

21:

Studies have demonstrated the use of statins in patients with previous stroke or transient ischemic attack (TIA), presumed to be of atherosclerotic origin, may reduce the risk of future ischemic stroke without increasing the risk of intracerebral hemorrhage (Kleindorfer et al., Stroke 2021, 52: e364-e467).

22:

Based on the results from the Stroke Prevention by Aggressive Reduction in Cholesterol Levels (SPARCL) trial, high-intensity statin therapy, when appropriate, is recommended for all patients with ischemic stroke and transient ischemic attack to reduce the risk of stroke and cardiovascular events (Grundy et al., Circulation 2019, 139: e1082-e143).

23:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

24:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

25:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

26:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

27:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor

patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

28:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

29:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

30:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen

- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

31:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

32:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

33:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

34:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week,

secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

35:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

36:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

37:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

38:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.